EDUSPARK EXAMS CONSULT

COMPUTING
Essay and Objective
1hour 45minutes

1&2

Name: _____

Index number: _____

Date: _____

B.E.C.E 2027 STEPPING STONE MOCK – SEPTEMBER, 2026

COMPUTING [100 marks]

*Write your **name** and **date** in **ink** in the spaces provided above*
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

- ✓ *This booklet consists of two papers, Paper **2** and paper **1***
- ✓ *Answer paper **2** in your answer booklet and paper **1** in the objective test answer sheet*
- ✓ *Paper **2** will last 1hour.*
- ✓ *Paper **1** will last* 45minutes.
- ✓ ***Do not start paper 1 until you are told to do so***

For Examiners Use Only	
Question Number	Marks
TOTAL	

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PAPER 2 - ESSAY [60 MARKS]

This paper is in two sections A and B Answer Question 1 in Section A and any other three questions in Section B. Answer all the questions in the spaces provided in this question paper. At the end of the examination, you should submit the entire question paper to the invigilator

SECTION A

[24 MARKS]

1. Your school has received ten new desktop computers for Basic 7 and 8. As computing prefect, you have been asked to set up the laboratory and prepare the students for the BECE.

(a) Three computers are not displaying any information on the screen after you switched them on.

(i) What could be the cause of this problem?

_____

_____

_____

____ [1 mark]

(ii) What two steps will you take to solve it?

_____

_____

_____

____ [2 marks]

(b) You noticed students sitting too close to the monitor and bending their backs during typing lessons.

(i) What two problems can this cause to their health?

_____

_____

_____

____ [2 marks]

(ii) What two corrections will you give them?

_____

_____

_____

____ [2 marks]

(c) You are to prepare the computer laboratory rules in Microsoft Word for 20 computers.

(i) Your document has many errors and you want to correct them quickly. How will you do it?

_____

_____

_____

____ [2 marks]

(ii) You want to save the file so that no student can edit it. What will you do?

_____

_____

_____

____ [2 marks]

(d) The teacher recorded examination scores in Microsoft Excel as follows: B2=67, C2=78, D2=82.
(i) The teacher wants to know the total score in E2. What will you write in E2?

_____

_____

_____

____ [2 marks]

(ii) The teacher also wants the smallest score in F2. What function will you use?

_____

_____

_____

____ [1 mark]

(e) The headteacher reports that students are downloading unknown files and the computers are now slow.
(i) What is likely to be the cause of the slowness?

_____

_____

_____

____ [1 mark]

(ii) How will you protect the computers from this problem in future?

_____

_____

____ [2 marks]

(f) The school wants to buy a machine that can clean the laboratory without being controlled by a person.
(i) What type of technology is this?

___ [1 mark]

(ii) Mention two other areas where this technology can be used in your community.

_____ [2 marks]

(iii) State one challenge of using this technology.

____. [1 mark]

(g) A Basic 7 student cannot understand how a computer works as a system.
(i) What will you tell him are the parts that make a complete working computer?

_____ [2 marks]

(ii) How do these parts work together?

____ [1 mark]

SECTION B
[36MARKS]

*Answer **three** questions **only** from this section.*

2. Your school has been connected to the internet for the first time.

(a) The PTA chairman asks you why the school needs internet. Give three reasons. [3 marks]

(b) You searched for "BCE" and got no results, but you wanted "BECE". How can you improve your search to get correct results? Give two methods. [2 marks]

(c) A student created a fake Facebook account to insult another student. What is this behaviour called? [1 mark]

(d) What two actions should the school take to stop this behaviour? [2 marks]

(e) The school email account was hacked after the password was shared. How should a good password be created? Give two characteristics. [2 marks]

(f) State two ways the school can keep its information safe online. [2 marks]

3. (a)i. Parallel computing is on the rise in this information era, briefly explain parallel computing to your friends. [3marks]

ii. State three types of parallel computing. [3marks]

(b) The hard disk and solid state drive are both non-volatile storage, state any two differences between the hard disk and the solid state drive. [2marks]

(c) In the process of compressing file for an email communication, States the two main techniques that can be used for compressing files. [2marks]

(d) Assistive and adaptive technologies helps individuals with disabilities to ease their lives in society these days, as a computing student, differentiate between assistive technologies and adaptive technologies. [2marks]

4. a. You are a secretary at the Ministry of Health in Accra, and you've been tasked with preparing a report on the national health insurance scheme. You've made some edits to the document, but you realize you need to revert to a previous version. How do you undo the edited changes in the word processing document? [3 marks]

b. The Korle Bu Teaching Hospital in Accra is considering introducing robots to improve healthcare services. Outline three potential uses of robots in the health sector. [3 marks]

c. You're a researcher at the University of Ghana, and you've been invited to present your findings on the impact of climate change on agriculture in Ghana at a conference in Kumasi. How can you use presentation software to enhance the effectiveness of your presentation and engage your audience? [3 marks]

d. How do you use the Slide Master to apply consistent formatting to slides? [3 marks]

5. a. What is a SmartArt? [2 marks]

b. You're a computer technician and a client has asked you to optimize their computer's storage. State three advantages of drive partitioning that you would explain to the client. [3 marks]

c. Give two functions of the network layer in the OSI Model. [4 marks]

d. You're setting up a new computer system for a small business. List three direct entry devices that you would recommend to the business owner to improve data entry efficiency. [3 marks]

PAPER 1 – OBJECTIVE TEST [40marks]

*Each question is followed by four options lettered **A** to **D**. Find out the correct option for each question and shade in pencil on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only **one** answer to each question*

1. The basic unit of information in computing is:
A. Byte
B. Bit
C. Kilobyte
D. Megabyte

2. What is the main function of an operating system?
A. To compile programs
B. To manage computer hardware and software resources
C. To facilitate internet browsing
D. To perform arithmetic operations

3. Which of the following is an example of application software?
A. Microsoft Word
B. Linux
C. BIOS
D. UNIX

4. What device is primarily used to capture video input?
A. Scanner
B. Microphone
C. Webcam
D. Printer

5. Which of the following protocols is used for secure internet transactions?
A. FTP
B. HTTP
C. HTTPS
D. SMTP

6. Which device is used for entering text into a computer?
A. Mouse
B. Keyboard
C. Monitor
D. Printer

7. RAM in computing stands for:
A. Read Access Memory

B. Random Access Memory
C. Read And Modify
D. Random And Modify

8. Which of the following is a volatile memory type?
A. ROM
B. SSD
C. RAM
D. HDD

9. The part of the computer responsible for executing instructions is the:
A. Hard drive
B. CPU
C. RAM
D. Motherboard

10. The shortcut key to copy text in most software applications is:
A. Ctrl + A
B. Ctrl + S
C. Ctrl + C
D. Ctrl + X

11. Which of the following is not an operating system?
A. Windows 10
B. MacOS
C. Linux
D. Microsoft Word

12. What does WAN stand for in networking?
A. Wide Area Network
B. World Area Network
C. Wireless Area Network
D. Web Area Network

13. Which of the following is a primary function of an antivirus program?
A. Increase processing speed
B. Detect and remove malware
C. Manage network connections
D. Optimize software performance

14. In computing, what does URL stand for?
A. Uniform Resource Locator
B. Universal Resource Locator
C. Uniform Retrieval Locator

D. Universal Retrieval Locator

15. Which component is considered the "brain" of the computer?
A. Hard drive
B. CPU
C. RAM
D. Motherboard

16. Which of the following is a programming language?
A. HTML
B. HTTP
C. USB
D. ISP

17. The main function of a router is to:
A. Store data
B. Print documents
C. Connect different networks
D. Perform calculations

18. Which of the following is a disadvantage of cloud computing?
A. Limited storage capacity
B. Requires internet connection
C. High hardware costs
D. Limited accessibility

19. The software that allows users to browse the internet is called:
A. Browser
B. Search engine
C. Web server
D. ISP

20. A small, portable device used to store and transfer data is called:
A. Hard drive
B. SSD
C. USB flash drive
D. CD-ROM

21. Which of the following is not a characteristic of a good password?
A. Short and simple
B. Combination of letters, numbers, and symbols
C. Unique and not easily guessable
D. Regularly updated

22. The device used to connect a computer to a telephone line for internet access is called:
A. Router
B. Switch
C. Modem
D. Hub

23. The process of starting or restarting a computer is known as:
A. Booting
B. Logging in
C. Installing
D. Formatting

24. The command to create a new folder in most operating systems is:
A. Ctrl + N
B. Ctrl + F
C. Ctrl + T
D. Ctrl + O

25. Which of the following is not an input device?
A. Scanner
B. Mouse
C. Printer
D. Keyboard

26. The software used to create spreadsheets is:
A. Microsoft Word
B. Microsoft Excel
C. Microsoft PowerPoint
D. Microsoft Access

27. Which of the following is an output device?
A. Monitor
B. Keyboard
C. Mouse
D. Scanner

28. Which type of storage device is known for having no moving parts?
A. HDD
B. SSD
C. Floppy disk
D. DVD

29. In networking, what does LAN stand for?

A. Local Access Network
B. Local Area Network
C. Large Area Network
D. Long Area Network

30. Which of the following is a use of database management systems?
A. Browsing the internet
B. Managing large sets of structured data
C. Editing images
D. Designing websites

31. The process of converting human-readable data into a coded form is known as:
A. Compression
B. Encryption
C. Decompression
D. Decryption

32. Which device is used to convert printed documents into digital format?
A. Printer
B. Monitor
C. Scanner
D. Keyboard

33. The term used for a malicious software designed to harm, exploit, or otherwise compromise the integrity of a computer system is:
A. Virus
B. Program
C. Software
D. Application

34. Which of the following is an example of an optical storage device?
A. USB flash drive
B. Hard disk
C. DVD
D. SSD

35. Which component of a computer is responsible for generating the visual output on a display screen?
A. Sound card
B. Network card
C. Graphics card
D. Motherboard

36. Which of the following is an example of an input/output device?

A. Printer
B. Monitor
C. Keyboard
D. Touchscreen

37. What is the primary purpose of using a firewall in a computer network?
A. To speed up the internet connection
B. To protect the network from unauthorized access
C. To store backup data
D. To monitor system performance

38. Which type of memory is used to store the BIOS settings in a computer?
A. RAM
B. ROM
C. Cache
D. Flash memory

39. The process of finding and correcting errors in software code is known as:
A. Compiling
B. Debugging
C. Executing
D. Testing

40. Which of the following is a common use of spreadsheets in business?
A. Writing reports
B. Analyzing financial data
C. Designing websites
D. Sending emails

EDUSPARK EXAMS CONSULT
BECE 2027 STEPPING STONE MOCK
COMPUTING
FINAL MARKING SCHEME

SECTION A (OBJECTIVE TEST) [40marks]

1. B	11. D	21. A	31. B
2. B	12. A	22. C	32. C
3. A	13. B	23. A	33. A
4. C	14. A	24. A	34. C
5. C	15. B	25. C	35. C
6. B	16. A	26. B	36. D
7. B	17. C	27. A	37. B
8. C	18. B	28. B	38. B
9. B	19. A	29. B	39. B
10. C	20. C	30. B	40. B

PAPER 2
SECTION A

QUESTION 1

(a) Three computers are not displaying any information.

(i) What could be the cause? [1 mark]

- Monitor not connected / VGA/HDMI cable loose
- Monitor not switched on / Power cable not connected
- Faulty monitor / Faulty graphics card
- Computer not switched on

[Any 1 point = 1 mark]

(ii) What two steps will you take to solve it? [2 marks]

- Check and connect the VGA/HDMI cable properly
- Switch on the monitor / Check power cable
- Test with another monitor
- Restart the computer
- Call technician if faulty

[1 mark each for any 2 = 2 marks]

(b) Students sitting too close and bending backs.

(i) Two problems to health [2 marks]

- Eye strain / Poor vision / Eye pain
- Backache / Waist pain / Poor posture
- Neck pain / Headache
- Fatigue / Tiredness
- Repetitive Strain Injury

[1 mark each for any 2 = 2 marks]

(ii) Two corrections [2 marks]

- Sit upright / Straight back / Use chair with back support
- Keep eyes at least 50cm from screen
- Keep feet flat on floor
- Take regular breaks / 20-20-20 rule
- Adjust chair and table height / Ergonomic sitting

[1 mark each for any 2 = 2 marks]

(c) Prepare lab rules in Word.

(i) Document has many errors, how to correct quickly? [2 marks]

- Use Spell Checker / Spelling and Grammar check
- Use AutoCorrect
- Use Thesaurus / Dictionary
- Use Find and Replace

[2 marks for correct method]

(ii) Save file so no student can edit [2 marks]

- Protect document with password / Encrypt with password
- Save as PDF / Read-only
- Use Restrict Editing feature
- Set file to Read-Only property

[Any 1 method well explained = 2 marks]

(d) Excel scores B2=67, C2=78, D2=82

(i) Total in E2 [2 marks]

- =B2+C2+D2
- =SUM(B2:D2)
- =SUM(B2,C2,D2)

[Correct formula = 2 marks, without = sign = 1 mark]

(ii) Smallest score in F2 [1 mark]

- =MIN(B2:D2)
- =MIN(B2,C2,D2)

[1 mark]

(e) Computers slow due to unknown files.

(i) Likely cause [1 mark]

- Virus infection / Malware attack
- Too many downloaded files filling storage

[1 mark]

(ii) How to protect in future [2 marks]

- Install and update antivirus software
- Scan pen drives before opening
- Avoid downloading from unknown sites
- Enable firewall
- Educate students not to download unknown files

[1 mark each for any 2 = 2 marks]

(f) Machine to clean lab without human control.

(i) Type of technology [1 mark]

Robotics / Robot / Robotic technology

[1 mark]

(ii) Two other areas in community [2 marks]

- Farming / Spraying crops / Harvesting
- Hospital / Surgery / Caring for patients
- Factory / Manufacturing / Packing goods
- Security / Surveillance
- Domestic work / Cooking / Washing
- Education / Teaching

[1 mark each for any 2 = 2 marks]

(iii) One challenge [1 mark]

- Expensive to buy / High cost
- Can replace human jobs / Unemployment
- Requires skilled persons to maintain
- Can malfunction / Needs electricity

[1 mark]

(g) Computer as a system.

(i) Parts that make complete working computer [2 marks]
- Hardware, Software, Data/Information, User/People, Procedure

[1 mark each for any 2 = 2 marks]

(ii) How do these parts work together? [1 mark]
- User gives data/instructions through hardware, software processes it and gives output
- All parts depend on each other to function / Interrelated

QUESTION 3.

(a) Three reasons why school needs internet [3 marks]
- Research and learning / Access information
- Online teaching and learning / E-learning
- Communication / Email
- Download educational resources
- Access to BECE past questions

[1 mark each for any 3 = 3 marks]

(b) Searched BCE instead of BECE - improve search [2 marks]
- Check spelling / Type correct keyword BECE
- Use quotation marks "BECE"
- Use more specific keywords e.g. "BECE past questions"
- Use suggested search / Auto-correct

[1 mark each for any 2 = 2 marks]

(c) Fake account to insult [1 mark]
- Cyberbullying / Impersonation / Cyber harassment

[1 mark]

(d) Two actions to stop [2 marks]
- Report account / Block account
- Report to teacher / Parents / Authorities
- Educate students on cyber ethics
- Ban phones / Regulate social media use

[1 mark each for any 2 = 2 marks]

(e) Good password characteristics [2 marks]
- Should be at least 8 characters long
- Mix of uppercase, lowercase, numbers and symbols
- Should not be personal name or birth date
- Should be hard to guess

[1 mark each for any 2 = 2 marks]

(f) Two ways to keep information safe [2 marks]
- Use strong passwords
- Do not share passwords
- Use antivirus and firewall
- Regular backup
- Enable two-factor authentication

[1 mark each for any 2 = 2 marks]

QUESTION 2

(a)i.

Parallel computing is a method where multiple processors or computers work together simultaneously to solve a problem faster by dividing tasks among them.

[3marks]

2(a)ii.
- Bit-level parallelism
- Instruction-level parallelism
- Data parallelism

[1mark each]

b(i)

- Hard disks use spinning magnetic platters, while SSDs use flash memory with no moving parts.
- SSDs are faster and more durable, while hard disks are generally cheaper and offer more storage capacity.

[1mark each]

c.

- Lossless compression
- Lossy compression

[1mark each]

d. **Assistive technologies** are tools designed specifically to help individuals with disabilities perform tasks they otherwise couldn't. while **Adaptive technologies** are modifications of existing tools to make them usable by individuals with disabilities.

[1mark]

QUESTION 4

a. **How to undo edited changes in a word processing document**

To undo edited changes in a word processing document, you can:
1. Press Ctrl + Z (Windows) or
2. Click on the "Undo" button in the toolbar or ribbon.
This will revert the document to its previous state, allowing you to undo one or more changes.

[3 marks]

b. **Uses of robots in the health sectors**.

- Robots also assist in surgery.
- Robotic carts may be seen moving through corridors carrying medical supplies
- They are used to disinfect patient rooms and operating suites.
- They work in laboratories to take samples, analyse and store them.
- Robotic medical assistants monitor patient vital statistics
- Robotic assistants enter information into patient electronic health records.

[Any 3 for 3 marks]

c. **How presentation software enhance the effectiveness of presentations**

Presentation software enhances the effectiveness of presentations by:

- Allowing for visually engaging and interactive content
- Enabling the use of multimedia elements, such as images, videos, and audio
- Providing tools for organizing and structuring content
- Facilitating communication and engagement with the audience **[3 marks]**

d. **How to use the Slide Master to apply consistent formatting to slides**

To use the Slide Master to apply consistent formatting to slides:
1. Go to the "View" tab in PowerPoint and select "Slide Master".
2. Make changes to the layout, design, and formatting of the Slide Master, such as adding logos, changing fonts, or adjusting margins.
3. Close the Slide Master view, and the changes will be applied to all slides in the presentation.
This ensures consistency in design and formatting throughout the presentation. **[3 marks]**

QUESTION 5

a. **What is a SmartArt?**

SmartArt is a tool in Microsoft Office applications (such as Word, PowerPoint, and Excel) that allows users to create professional-looking diagrams, charts, and graphics from text-based information.

[2 marks]

b. **Advantages of drive partitioning**.

- Better organization of data
- Protects important files from corruption
- Allows multiple operating systems
- Easier backup and restore
- Improves performance
- Easy reinstallation of OS without data loss
- Better disk space management **[Any 3 for 3 marks]**

c. **Functions of the network layer in the OSI Model**.

- Routing: Determines the best path for data to travel between networks.
- Logical Addressing: Assigns logical addresses (IP addresses) to devices on a network.
- Packet Forwarding: Forwards packets of data between networks.

- Fragmentation and Reassembly: Breaks down large packets into smaller ones and reassembles them at the receiving end.
- Congestion Control: Helps prevent network congestion by managing traffic flow.
- Error Handling: Reports errors and provides feedback to the transport layer.
- Path Selection: Selects the best path for data to travel based on network conditions.

[Any 2 for 4 marks]

d. **Direct entry devices**.

- Keyboard
- Scanner
- Barcode Reader
- Smart Card Reader
- Biometric Devices (e.g., Fingerprint Reader, Facial Recognition)
- OCR (Optical Character Recognition) Scanner
- Camera (e.g., Webcam, Document Camera)
- Microphone
- Graphics Tablet
- Touchscreen **[Any 3 for 3 marks]**